Project Documentation

Project Title: Product Inventory Management System

1. Introduction

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2. Project Overview

Purpose: The system helps users manage product inventory using a secure backend built with Node.js, Express.js, and MongoDB.

Features:

- User registration and login using JWT authentication
- Add, view, update, and delete products (CRUD)
- Secure API routes with authentication middleware
- Product search functionality
- Each user can manage only their own products

3. Architecture

Frontend: React.js with Bootstrap / Material UI

Backend: Node.js and Express.js for server logic and API development

Database: MongoDB for storing users and product information

4. Setup Instructions

Prerequisites:

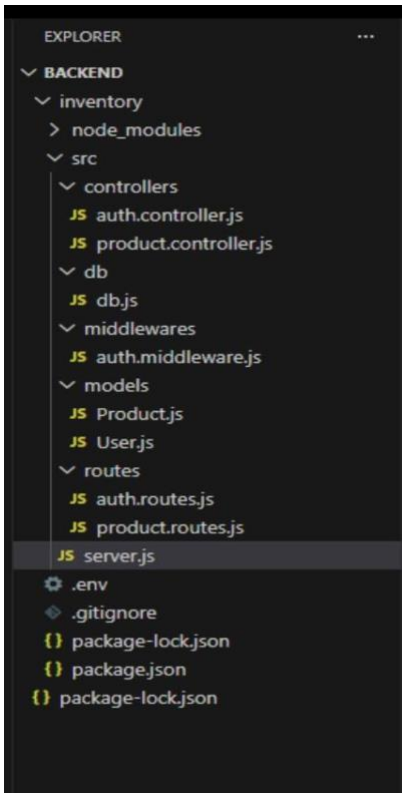
- Node.js
- MongoDB
- Git
- React.js
- Visual Studio Code

Installation Steps:

1. Clone the repository using git clone
2. Install client dependencies: cd client -> npm install
3. Install server dependencies: cd server -> npm install

5. Folder Structure

```
SB-Works/
 ├── client/       (React frontend)
 │   ├── components/
 │   └── pages/
 └── server/       (Node.js backend)
     ├── routes/
     ├── models/
     └── controllers/
```



6. Running the Application

cd inventory
npm install
npm start
npm run dev
PS C:\Users\acer\Desktop\backend\inventory>

Access the application at: http://localhost:5000

7. API Documentation

User APIs:

- /api/user/register
- /api/user/login

Project APIs:

- /api/projects/create
- /api/projects/:id
- /api/apply

Chat APIs:

- /api/chat/send
- /api/chat/userId

8. Authentication

JWT-based authentication is used to provide secure login for users. Middleware verifies the token and protects private API routes from unauthorized access.

9. User Interface

- Landing Page
- Freelancer Dashboard
- Admin Panel
- Project Details Page

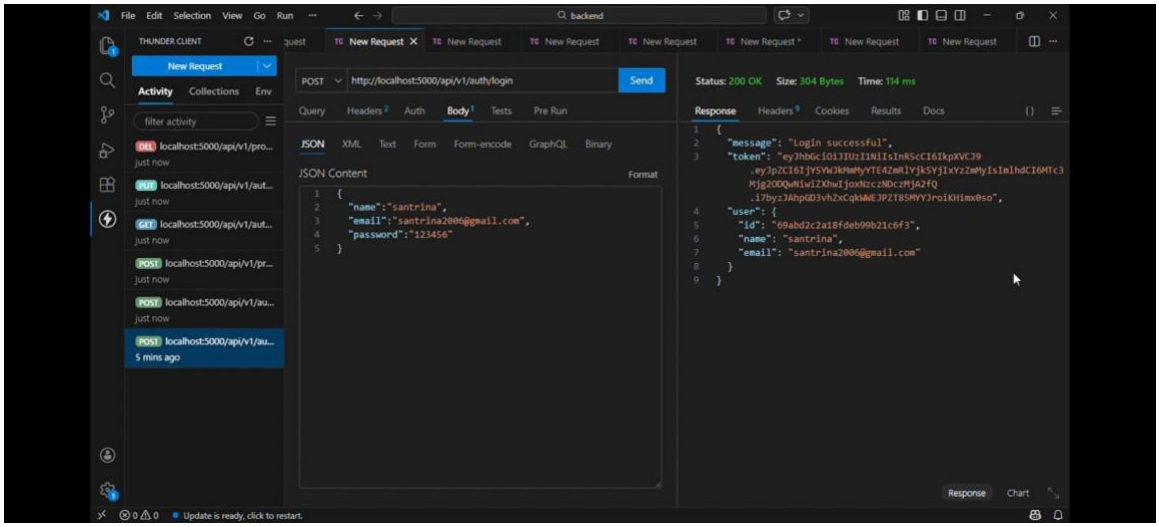
The system provides simple and user-friendly pages such as Landing Page, Freelancer Dashboard, Admin Panel, and Project Details Page. These pages help users easily interact with the application.

10. Testing

Manual testing during development milestones. The application is tested manually during development to ensure all features work properly. Tools like Postman and Chrome Developer Tools are used to test API responses and functionality.

Tools Used: Postman, Chrome Developer Tools

11. Screenshots

Screenshots of the working application and API testing results are included in this section

Final output: